

Souptik Sinha

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Education

Illinois institute of technology

Aug 2024 - Jun 2026

Master of Science in Computer Science

- **Coursework:** Machine Learning, Database Organization, Computer Networks, Data Structures and Algorithms, Generative AI, Software Engineering, System Design

Work Experience

CXDS

Jun 2024 - Aug 2024

AI software Engineer

India

- Built CNN-based fashion recommendation engine using ResNet18 for visual feature extraction and weighted embedding aggregation, achieving 92-98% accuracy with <3ms inference time.
- Developed hybrid KNN+LightGBM ensemble model with feature engineering for style prediction from user ratings
- Designed and deployed FastAPI backend with PostgreSQL database integration for real-time style predictions, implementing RESTful endpoints for moodboard rating collection, model inference pipeline, and user profile management with sub-100ms response times

Self-Employed

May 2023 - Jul 2024

Fullstack Developer

Remote-based Clients

- Delivered 5+ full-stack web applications using React.js, Node.js, and PostgreSQL with modular architecture and automated testing (Jest, Mocha), improving reliability and cutting post-deployment bugs by 35%.
- Implemented secure JWT/OAuth 2.0 authentication and payment systems (Stripe, PayPal) with AES-256/TLS encryption, decreasing checkout abandonment by 30% and raising conversion by 25%.
- Decreased REST API latency through Redis caching, connection pooling, and AWS ELB, sustaining 99.9% uptime.
- Automated CI/CD pipelines on GitHub Actions and AWS deployments, reducing release cycle time by 70%.

Projects

AI-Powered Learning Platform

Jun 2025

- Built and deployed an AI-powered learning app (Next.js, React, Vapi) delivering 5+ reinforcement-learning-based voice tutors, supporting 1K+ sessions/month with 98% uptime.
- Created an accessible UI (TailwindCSS) and engineered full-stack capabilities (API routes, WebSockets for real-time updates). Integrated Clerk for authentication and Supabase (100K+ users / 500GB data); deployed on Vercel.

Football Prediction -ML

Aug 2025

- Constructed and validated a Random Forest Classifier for football match outcome prediction with 22 features, achieving a top 5 leaderboard ranking and validating performance via cross-validation (F1, precision, recall).
- Built Flask REST APIs (Scikit-learn, Pandas) for real-time predictions and migrated the models to AWS (S3, Lambda) using joblib persistence, cutting experimentation time by 40%.
- Developed a React/Tailwind frontend for real-time match insights and analytics, supporting 2K+ users with <500ms latency.

JoinAHACK- Full-Stack Social Media Application

Jun 2025

- Launched a social media platform with React.js/Vite frontend and Node.js/Express backend, enabling students to showcase projects, find hackathon teammates, and organizers to manage events with real-time messaging.
- Integrated secure JWT authentication(2 Factor), Postgres on Supabase database management, and real-time updates, to solve student project visibility and hackathon team formation.
- Scalable to 3K+ users and 500+ project submissions across 20+ events.

Skills

- **Programming Languages:** JavaScript (ES6+), Python, TypeScript, SQL, HTML5, CSS3
- **Frameworks & Libraries:** React.js, Next.js, Tailwind CSS, Node.js, TensorFlow, Scikit-learn, Flask
- **Databases:** PostgreSQL, MySQL, Supabase, MongoDB
- **Machine Learning/AI:** PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Matplotlib, Hugging Face Transformers, Playwright-, Cross-Validation, RAG, Generative AI, LangGraph
- **Cloud & DevOps:** Git, Github, Docker, AWS(Lambda, EC2, S3, Elastic Beanstalk), Vercel, CI/CD, JWT Authentication, API Testing, Agile Development, API, Google Cloud Platform (GCP), Firebase, Data Processing, Web Scraping, Prompt Engineering, GitHub, Heroku
- **Testing & Automation:** Jest , Workflow Automation, API Testing, Playwright, Mocha